



Maria Teleki

SOCIALLY ROBUST AI

PhD Student in Computer Science at **Texas A&M University** in *College Sta., TX*
Student Researcher at **Google DeepMind** in *Cambridge, MA*

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SUMMARY

I build methods, benchmarks, and audits that improve AI robustness in socially complex situations.
My research spans **Speech AI** and **Computational Social Science**, across three threads:

Disfluency-Aware AI: *Is AI robust to how people naturally speak?*

→ Preprint, Preprint, ICASSP '26, INTERSPEECH '25, INTERSPEECH '24, LREC-COLING '24

Personalized AI: *Is AI robust to each person's uniqueness?*

→ ICWSM '25, INTERSPEECH '26, ACL '26

Governable AI: *Is AI robust within the sociopolitical structures it operates in?*

→ FAccT '26

EDUCATION

2022 – Present | PhD Computer Science at Texas A&M University

Advised by James Caverlee.

2017 – 2022 | B.S. Computer Science at Texas A&M University

Summa Cum Laude. Emphasis in Electrical Engineering. Minor in Math.

PUBLICATIONS

- P1. **The Due Process Deficit: Auditing AI Governance in U.S. Higher Education**
Maria Teleki, Anna Seo Gyeong Choi, Anne Duray, Haoran Liu, Julie Zhang, Xiangjue Dong, Dilma Da Silva, Allison Koenecke, and James Caverlee
– *Collab. with Cornell, AAAS Science and Technology Policy Fellow*
FAccT '26 The ACM Conference on Fairness, Accountability, and Transparency [Link]
→ Presented at: IC2S2 Parallel Talk '26 and The Contentious Politics of AI Network (Remote)
- P2. **Z-Scores: A Metric for Linguistically Assessing Disfluency Removal**
Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, and James Caverlee
ICASSP '26 IEEE International Conference on Acoustics, Speech and Signal Processing [Link]
- P3. **The Voice Behind the Words: Quantifying Intersectional Bias in SpeechLLMs**
Shree Harsha Bokkahalli Satish, Christoph Minixhofer, **Maria Teleki**, James Caverlee, Ondrej Klejch, Peter Bell, Gustav Eje Henter, and Éva Székely
– *Collab. with KTH, University of Edinburgh*
INTERSPEECH '26 Conference of the International Speech Communication Association [Link]
- P4. **CHOIR: Collaborative Harmonization fOr Inference Robustness**
Xiangjue Dong, Cong Wang, **Maria Teleki**, Millennium Bismay, Ruihong Huang, and James Caverlee
ACL '26 The Annual Meeting of the Association for Computational Linguistics [Link]
- P5. **I want a horror – comedy – movie: Slips-of-the-Tongue Impact Conversational Recommender System Performance**

Maria Teleki, Lingfeng Shi, Chengkai Liu, and James Caverlee
INTERSPEECH '25 Conference of the International Speech Communication Association [Link]

- P6. **Masculine Defaults via Gendered Discourse in Podcasts and Large Language Models**
Maria Teleki, Xiangjue Dong, Haoran Liu, and James Caverlee
ICWSM '25 The International AAAI Conference on Web and Social Media [Link]
→ Presented at: *IC2S2 Parallel Talk '25 and SIcon@ACL '25*
→ Media Coverage: *MASKulinity Podcast*
- P7. **A Survey on LLMs for Story Generation**
Maria Teleki, Vedangi Bengali, Xiangjue Dong, Sai Tejas Janjur, Haoran Liu, Tian Liu, Cong Wang, Ting Liu, Yin Zhang, Frank Shipman, and James Caverlee
EMNLP Findings '25 Findings of the Association for Computational Linguistics: EMNLP 2025 [Link]
- P8. **Comparing ASR Systems in the Context of Speech Disfluencies**
Maria Teleki, Xiangjue Dong, Soohwan Kim, and James Caverlee
INTERSPEECH '24 Conference of the International Speech Communication Association [Link]
- P9. **Quantifying the Impact of Disfluency on Spoken Content Summarization**
Maria Teleki, Xiangjue Dong, and James Caverlee
LREC-COLING '24 The Joint International Conference on Computational Linguistics, Language Resources and Evaluation [Link]
- P10. **Co2PT: Mitigating Bias in Pre-trained Language Models through Counterfactual Contrastive Prompt Tuning**
Xiangjue Dong, Ziwei Zhu, Zhuoer Wang, **Maria Teleki**, and James Caverlee
— Collab. with George Mason University
EMNLP Findings '23 Findings of the Association for Computational Linguistics: EMNLP 2023 [Link]

PREPRINTS

- R1. **A Cross-Community Agenda for Speech AI**
Maria Teleki^{*}, Kimi V. Wenzel^{*}, Anna Seo Gyeong Choi^{*}, Tobias Weinberg^{*}, Shree Harsha Bokkiahalli Satish^{*}, Stephanny Sanchez^{*}, Belu Ticona^{*}, Ariadna Sanchez^{*}, Yash Sonkar^{*}, Aarti Mathur^{*}, Christoph Minixhofer^{*}, Abraham Glasser^{*}, Raja Kushalnagar^{*}, James Caverlee^{*}, Minha Lee^{*}, Shaomei Wu^{*}, Alyssa Hillary Zisk^{*}, Éva Székely^{*}, Dylan Gaines^{*}, Angelika Seeschaaf Veres^{*}, Seray Ibrahim^{*}, Nicholas Cummins^{*}, and Allison Koenecke^{*}
Under Submission at ARR ACL Rolling Review [Link]
- R2. **The Role of Disfluencies in Speech Translation**
Maike Züfle^{*}, **Maria Teleki**^{*}, Vilém Zouhar, Fabian Retkowski, Oliver Grabner, Alex Waibel, Jan Niehues, and James Caverlee
— Collab. with KIT, ETH Zürich
Under Submission at TACL Transactions of the Association for Computational Linguistics [Link]
- R3. **SpeechSpectrum: A Framework for Speech-to-Text Representation Along the Linguistic Fidelity Spectrum**
Anna Seo Gyeong Choi^{*}, **Maria Teleki**^{*}, Miguel del Rio, James Caverlee, Corey Miller, and Allison Koenecke
— Collab. with Cornell, Rev AI
In Prep. for CSCW The ACM Conference on Computer-Supported Cooperative Work and Social Computing [Link]
→ Presented at: *IC2S2 Poster '26*

- R4. **Beyond Single Ground Truth: Reference Monism as Epistemic Injustice in ASR Evaluation**
 Anna Seo Gyeong Choi, **Maria Teleki**, James Caverlee, Miguel del Rio, Corey Miller, and Hoon Choi
 — Collab. with Cornell, Rev AI, Kangwon National University
In Prep. for ARR ACL Rolling Review [Link]
 → Presented at: *IC2S2 Poster ‘26*
- R5. **Conversational Speech Reveals Systematic Robustness Failures in LLMs**
Maria Teleki, Sai Janjur, Haoran Liu, Oliver Grabner, Ketan Verma, Thomas Docog, Xiangjue Dong, Lingfeng Shi, Cong Wang, Stephanie Birkelbach, Jason Kim, Yin Zhang, Éva Székely, and James Caverlee
 — Collab. with KTH
In Prep. for ARR The ACL Rolling Review [Link]

POSTERS, DEMOS, TECHNICAL REPORTS, & MENTEE PAPERS

- M1. **Walk a Mile in My Voice: Voice Conversion Shapes Trust, Attribution, and Empathy in Human–AI Speech Interactions**
 Shree Harsha Bokkahalli Satish, **Maria Teleki**, Christoph Minixhofer, Ondrej Klejch, Peter Bell, and Éva Székely
 — Collab. with KTH, University of Edinburgh
IUI ‘26 (Poster) The ACM Conference on Intelligent User Interfaces [Link]
 → Presented at: *CHI ‘26 (Poster)*
- M2. **UG 1ST AUTHOR**
PromptHelper: A Prompt Recommender System for Encouraging Creativity in AI Chatbot Interactions
 Jason Kim, **Maria Teleki**, and James Caverlee
CHI ‘26 (Poster) The ACM CHI Conference on Human Factors in Computing Systems [Link]
 → Presented at: *IC2S2 Poster ‘26*
- M3. **UG 1ST AUTHOR**
SocialPulse: An Open-Source Subreddit Sensemaking Toolkit
 Stephanie Birkelbach, **Maria Teleki**, Nehul Bhatnagar, Peter Carragher, Xiangjue Dong, and James Caverlee
 — Collab. with CMU, Revionics
SocialLLM @ ICWSM ‘26 The Workshop on Large Language Models for Social Reasoning and Simulation at the International AAAI Conference on Web and Social Media [Link]
 → Presented at: *IC2S2 Parallel Talk ‘26*
- M4. **UG 1ST AUTHOR**
PodChecker: An Interpretable Fact-Checking Companion for Podcasts
 Anna Irmetova, Haoran Liu, **Maria Teleki**, Peter Carragher, and James Caverlee
 — Collab. with CMU
MisD @ ICWSM ‘26 The Workshop on Misinformation Detection in the Era of LLMs at the International AAAI Conference on Web and Social Media [Link]
- M5. **UG 1ST AUTHOR**
Detecting and Mitigating Demographic Bias in LLM-Based Resume Evaluation
 Oluwadayo Bamgbelu, **Maria Teleki***, Xiangjue Dong*, and James Caverlee
In Preparation for ARR (Short) ACL Rolling Review [Link]
 → Presented at: *IC2S2 Poster ‘26*
- M6. **MS 1ST AUTHOR**

When Behavioral Traces Mislead: Evaluating Evidence Adequacy in Learning Analytics

Junhyeon Kwon, Jobin Varughese, **Maria Teleki**, and Mohsen Dorodchi

— Collab. with UNC Charlotte

In Prep. for LAK (Short) *The International Learning Analytics & Knowledge Conference* [Link]

→ Presented at: *IC2S2 Parallel Talk '26*

- M7. **AI in the Loop: A Multimodal Assistant for Real-Time Lecture Comprehension**
Amanda Lacy, **Maria Teleki**, Esau Hutcherson, Jobin Varughese, Jun Kwon, Frank Shipman, and Tracy Hammond
Learning at Scale Demo '26 The ACM Learning @ Scale Conference [Link]
- M8. **A Survey on LLM Inference-Time Self-Improvement**
Xiangjue Dong*, **Maria Teleki***, and James Caverlee
arXiv [Link]
- M9. **MS 1ST AUTHOR**
DACL: Disfluency Augmented Curriculum Learning for Fluent Text Generation
Rohan Chaudhury, **Maria Teleki**, Xiangjue Dong, and James Caverlee
LREC-COLING '24 The Joint International Conference on Computational Linguistics, Language Resources and Evaluation [Link]
- M10. **Howdy Y'all: An Alexa TaskBot**
Majid Alfifi, Xiangjue Dong, Timo Feldman, Allen Lin, Karthic Madanagopal, Aditya Pethe, **Maria Teleki**, Zhuoer Wang, Ziwei Zhu, and James Caverlee
Alexa Prize TaskBot Challenge Proceedings 2022 Alexa Prize TaskBot Challenge [Link]

EXPERIENCE

Google DeepMind | Student Researcher | Cambridge, MA | June – Sep '26

Working on Speech AI + CSS project on the Frontier AI team.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '21

BERT embeddings and PCA to surface which semantic dimensions of coupon content predicted CTR.

The Hi, How Are You Project | Volunteer | Austin, TX | May – Dec '20

Built the “Friendly Frog” Alexa Skill with the organization at the beginning of the COVID-19 pandemic to promote mental health.

RetailMeNot | Software Engineering Intern | Austin, TX | May – August '20

Built the “RetailMeNot DealFinder” Alexa Skill to help users activate cash back offers.

Silicon Labs | Applications Engineering Intern | Austin, TX | May – August '19

Built a Python library adopted by customers for real-time hardware bus traffic analysis across three protocols.

SERVICE

Workshop Organizer: **Speech AI for All: The What, How, and Who of Measurement**

Kimi V. Wenzel, Alisha Pradhan, **Maria Teleki**, Tobias Weinberg, Robin Netzorg, Alyssa Hillary Zisk, Anna Seo Gyeong Choi, Jingjin Li, Raja Kushalnagar, Colin Lea, Abraham Glasser, Christian Vogler, Nan Bernstein Ratner, Ly Xīnzhèn M. Zhāngsūn Brown, Allison Koenecke, Karen Nakamura, and Shaomei Wu

CHI '26 Workshop The ACM CHI Conference on Human Factors in Computing Systems [Link]

Program Committee (Reviewer) for **INTERSPEECH**: '26

Program Committee (Reviewer) for **IC2S2**: '26

Program Committee (Reviewer) for **ACL ARR**: Aug '24, Oct '24, Dec '24, May '25, Oct '25, Jan '26

Program Committee (Reviewer) for **ICWSM**: Jan '24, May '24, Sep '24, Jan '25, Sep '25, Jan '26

TEACHING

Certifications

2025 | Professional Development Mastery Certificate in Instruction & Assessment | GRAD Aggies, CIRTL@TAMU
Completed multi-semester program in evidence-based teaching, including syllabus design, learning assessment, and inclusive pedagogy.

Course Development

Spring 2026 | Course Developer | Texas A&M University, CSCE 676 Data Mining (Graduate Level)

Developed assignments using coding agents to test student implementations and video-recorded technical interviews that trained students to state their assumptions on vague questions.

Spring 2026 | Course Developer | Texas A&M University, CSCE 289 GenAI for Engineers (Undergraduate Level)

Helped shape the LLM and lightweight-coding curriculum for AI-driven engineering. Built the progression from foundational concepts through hands-on API work to critical evaluation, so students leave able to use these tools competently and skeptically.

Teaching Assistantships

Spring 2026 | Teaching Assistant | Texas A&M University, CSCE 676 Data Mining & Analysis

Guest Lectures

Fall 2025 | Speech Language Models | Texas A&M University, CSCE 676 Data Mining & Analysis

Spring 2025 | IR Evaluation and Learning to Rank | Texas A&M University, CSCE 670 Information Storage & Retrieval

Earlier Teaching

Dec 2018 – Dec 2019 | Peer Teacher | Texas A&M University

Taught weekly review sessions and assisted TA at lab sessions for CSCE 121.

Sep 2016 – July 2017 | Afterschool Instructor | The Y (YMCA)

Taught weekly classes at local elementary schools and developed STEM curriculum: authored robotics & science instruction manuals for the Williamson County YMCA Afterschool program.

May 2016 – Aug 2016 | Camp Counselor | Camp Twin Lakes, The Y (YMCA)

Worked as a kindergarten camp counselor, and taught robotics specialty class in the afternoon for K–8th grade students.

MENTORING

★ student was an author on a published paper; ♠ student was an author on a Preprint paper; ▲ student completed their thesis; ◆ student received course credit (i.e. CSCE 485, CSCE 691); ♣ student had no publications prior to mentorship.

MS Students

Rohan Chaudhury [★▲] M.S. Computer Science – First Employment at Amazon

1st author

DACL: Disfluency Augmented Curriculum Learning for Fluent Text Generation *LREC-COLING '24*

Sai Tejas Janjur [★♠♣] M.S. Computer Science – 🏆 1st Place Award at Student Research Week, First Employment at NVIDIA

co-author

A Survey on LLMs for Story Generation *EMNLP Findings '25*

co-author

Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP '26*

co-author

Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Cong Wang [★♠♣] *M.S. Statistics* – Presented at Texas NLP Symposium

- co-author CHOIR: Collaborative Harmonization fOr Inference Robustness *ACL* '26
- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Junhyeon Kwon [♠♣] *M.S. Computer Science*

- 1st author When Behavioral Traces Mislead: Evaluating Evidence Adequacy in Learning Analytics *In Prep. for LAK (Short)*
- co-author AI in the Loop: A Multimodal Assistant for Real-Time Lecture Comprehension *Learning at Scale Demo* '26

Undergraduate Students

Soohwan Kim [★♠♣] *B.S. Computer Science* – First Employment at UPS

- co-author Comparing ASR Systems in the Context of Speech Disfluencies *INTERSPEECH* '24

Oliver Grabner [★♠♣] *B.S. Computer Science* – Presented at Texas NLP Symposium & Student Research Week, Intern at Samsung, Intern at Google

- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*
- co-author The Role of Disfluencies in Speech Translation *Under Submission at TACL*

Thomas Docog [★♠♣] *B.S. Computer Science* – Presented at Texas NLP Symposium, Intern at Ramirez

- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Ketan Verma [★♠♣] *B.S. Computer Science* – Intern at Samsung, M.S. at Cornell Tech

- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Stephanie Birkelbach [★♠♣♣] *B.S. Computer Science + B.S. Statistics* – REU at Rochester Institute of Technology w/ Ashique KhudaBukhsh

- 1st author SocialPulse: An Open-Source Subreddit Sensemaking Toolkit *SocialLLM @ ICWSM* '26
- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Jason Kim [★♠♣] *B.S. Computer Science* – Presented at Texas NLP Symposium, 🏆 1st Place Award at Student Research Week

- 1st author PromptHelper: A Prompt Recommender System for Encouraging Creativity in AI Chatbot Interactions *CHI* '26 (Poster)
- co-author Z-Scores: A Metric for Linguistically Assessing Disfluency Removal *ICASSP* '26
- co-author Conversational Speech Reveals Systematic Robustness Failures in LLMs *In Prep. for ARR*

Anna Irmetova [★♣♠] *B.S. Computer Science* – Presented at Student Research Week, Intern at Target

- 1st author PodChecker: An Interpretable Fact-Checking Companion for Podcasts *MisD @ ICWSM* '26

Oluwadayo (Dayo) Bamgbelu [♣] *B.S. Computer Science* – Presented at Texas NLP Symposium, Intern at Meta

- 1st author Detecting and Mitigating Demographic Bias in LLM-Based Resume Evaluation *In Preparation for ARR (Short)*

AWARDS

2022–2026 | Dr. Dionel Avilés '53 and Dr. James Johnson '67 Fellowship in Computer Science and Engineering
Multi-year fellowship supporting doctoral research in Computer Science.

2024, 2025, 2026 | Department of Computer Science & Engineering Travel Grant

2024 | [CRA-WP Grad Cohort for Women](#)

Selected participant; supported with travel support.

2017–2021 | President's Endowed Scholarship

Four-year scholarship supporting undergraduate studies.

2018 | Bertha & Samuel Martin Scholarship

INVITED TALKS

Fall 2026 | *Socially Robust AI* | [George Mason University](#) | Remote

Fall 2026 | *The Due Process Deficit* | [The Contentious Politics of AI](#) | Remote

Spring 2026 | *Accountable AI* | IPT Global | Houston, Texas

Fall 2025 | *Conversational AI* | [Revionics](#) | Austin, Texas

Spring 2025 | *How does ChatGPT work? & My research!* | [Texas A&M University, Club of Aggie Female Engineers \(C.A.F.E.\)](#) | College Station, Texas

Fall 2024 | *The Other AI: An Intuitive Understanding of Artificial Intelligence* | [Texas Tech University, School of Veterinary Medicine VBMA](#) | Amarillo, Texas

REFERENCES

James Caverlee

Professor, Department of Computer Science & Engineering at Texas A&M University

Tenneco Professor, College of Engineering at Texas A&M University

caverlee@tamu.edu

Dilma Da Silva

Regents Professor, Department of Computer Science & Engineering at Texas A&M University

dilma@cse.tamu.edu

Allison Koenecke

Assistant Professor, Department of Information Science at Cornell Tech

Field Faculty Member, Department of Computer Science at Cornell Tech

koenecke@cornell.edu

Éva Székely

Assistant Professor, Division of Speech, Music and Hearing at KTH Royal Institute of Technology, Sweden

szekely@kth.se